

are the conditions under which it will have a solution of a particular character; in fact, we shall assume without proof the following fundamental theorem of differential equation, viz.,

An ordinary differential equation of order n has a solution involving n independent arbitrary constants, and this solution is unique.

15.4. Illustrative Examples.

Ex. 1. Find the differential equation of all straight lines passing through the origin. [C. P. 2002]

$$\text{Let } y = mx$$

$$\dots (1)$$

be the equation of any straight line passing through the origin.

$$\text{Differentiating (1), } \frac{dy}{dx} = m.$$

$$\dots (2)$$

Eliminating m between (1) and (2), we get

$$y = x \frac{dy}{dx}, \text{ the required differential equation.}$$

Ex. 2. Find the differential equation form the relation.

$$x = a \cos t + b \sin t,$$

a and b being arbitrary constants.

Differentiating the given relation twice with respect to t , we get

and

$$x_2 = -a \cos t - b \sin t = -(a \cos t + b \sin t) = -x.$$

$$\therefore x_2 + x = 0 \text{ i.e., } \frac{d^2 x}{dt^2} + x = 0 \text{ is the required differential equation.}$$

Ex. 3. Eliminate a and b from $y = a \tan^{-1} x + b$.

Differentiating the given relation with respect to x ,

$$y_1 = \frac{a}{1+x^2}.$$

$$\therefore (1+x^2) y_1 = a.$$

$$\text{Differentiating, } (1+x^2) y_2 + 2xy_1 = 0.$$

This is the required eliminant.

15.5. Miscellaneous Workedout Examples.

Ex. 1. State the order and degree of the following differential equations.

$$(i) \frac{d^2 y}{dx^2} + 4 \left(\frac{dy}{dx} \right)^5 + 10y = 0$$

[C. P. 1998]

$$(ii) \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{2}{3}} = \frac{d^2 y}{dx^2}$$

[C. P. 1998]

$$(iii) y \left(\frac{dy}{dx} \right)^2 + 2y \frac{dy}{dx} - y = 0$$

[C. P. 1998]

$$(iv) \sqrt{1 + \left(\frac{dy}{dx} \right)^2} = 1 + x$$

[C. P. 1998]

Solution : (i) Here, order of the given differential equation is 2 and degree of the equation is 1.

$$(ii) \text{ Here, } \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^{\frac{2}{3}} = \frac{d^2 y}{dx^2}$$

$$\text{or, } \left[1 + \left(\frac{dy}{dx} \right)^2 \right]^2 = \left(\frac{d^2 y}{dx^2} \right)^3$$

Order of the differential equation is 2, while the degree of the equation is 3.

$$(iii) y \left(\frac{dy}{dx} \right)^2 + 2x \frac{dy}{dx} - y = 0.$$

Here, order of the differential equation is 1, and the degree of the equation is 2.

$$(iv) \sqrt{1 + \left(\frac{dy}{dx} \right)^2} = 1 + x$$

$$\text{or, } 1 + \left(\frac{dy}{dx} \right)^2 = (1 + x)^2$$

Here, order of the differential equation is 1 and degree is 2.

Ex. 2. Find the differential equation of which each of the following functions is the general solution (A, B being arbitrary constants) :

(i) $y = A \sin x + B \cos x$ [C. P. 1983, 88]

(ii) $y = e^{-x} (A \cos x + B \sin x)$ [C. P. 1996]

(iii) $y = A \cos (px - B)$, p being a parameter. [C. P. 2000]

(iv) $y = kx + k - k^3$, k being a parameter. [C. P. 1999]

(v) $y = A + \frac{B}{x}$, [B. P. 1988, C. P. '81, '94, '88]

(vi) $y = Ae^{2x} + Be^{-2x}$ [B. P. 2000]

Solution : (i) $y = A \sin x + B \cos x$... (1)

or, $\frac{dy}{dx} = A \cos x - B \sin x$

or, $\frac{d^2 y}{dx^2} = -A \sin x - B \cos x = -y$ [from (1)]

or, $\frac{d^2 y}{dx^2} + y = 0$ is the required differential equation.

(ii) $y = e^{-x} (A \cos x + B \sin x)$

or, $ye^x = A \cos x + B \sin x$... (1)

Differentiating w.r.t. x ,

$e^x \cdot y + e^x \cdot \frac{dy}{dx} = -A \sin x - B \cos x$

Differentiating again w.r.t. x ,

$e^{-x} \cdot y + e^x \cdot \frac{dy}{dx} + e^x \cdot \frac{dy}{dx} + e^x \cdot \frac{d^2 y}{dx^2} = -A \cos x + B \sin x$

or, $e^x \left(\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + y \right) = -ye^x$ [using (1)]

or, $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 2y = 0$ [$\because e^x \neq 0$]

This is the required differential equation.

$$(iii) y = A \cos(px - B)$$

$$\therefore \frac{dy}{dx} = -Ap \sin(px - B)$$

$$\text{and } \frac{d^2y}{dx^2} = -Ap^2 \cos(px - B) = -p^2 y$$

$$\therefore \frac{d^2y}{dx^2} + p^2 y = 0 \text{ is the required differential equation.}$$

$$(iv) y = kx + k - k^3$$

... (1)

$$\text{or, } \frac{dy}{dx} = k$$

Putting this value of k in (1), we get

$$y = x \cdot \frac{dy}{dx} + \frac{dy}{dx} - \left(\frac{dy}{dx} \right)^3$$

$$\text{or, } y = (1+x) \frac{dy}{dx} - \left(\frac{dy}{dx} \right)^3 \text{ is the required differential equation.}$$

$$(v) y = A + \frac{B}{x}$$

$$\text{or, } \frac{dy}{dx} = -\frac{B}{x^2}$$

$$\text{or, } x^2 \frac{dy}{dx} = -B$$

$$\text{Differentiating again, } 2x \frac{dy}{dx} + x^2 \frac{d^2y}{dx^2} = 0$$

$$\text{or, } \frac{d^2y}{dx^2} + \frac{2}{x} \frac{dy}{dx} = 0 \text{ is the required differential equation.}$$

$$(iv) y = Ae^{2x} + Be^{-2x}$$

$$\text{or, } \frac{dy}{dx} = 2Ae^{2x} + Be^{-2x}$$

$$\text{and } \frac{d^2y}{dx^2} = 4Ae^{2x} + 4Be^{-2x} = 4y$$

$$\text{i.e., } \frac{d^2y}{dx^2} - 4y = 0 \text{ is the required differential equation.}$$

Ex. 3. Find the differential equation of :

(i) All parabolas having their axes parallel to the y-axis.

[C. P. 1994, 95]

(ii) All circles passing through the origin and having their centres on the x-axis.

[B. P. '95, C. P. '99]

(iii) All the circles having a constant radius.

[C. P. '96, '97]

(iv) The system of circles having a constant radius a and having their centres on the x-axis.

[C. P. 2000]

Solution : (i) Equation of any parabola whose axis is parallel to the x-axis is $y = Ax^2 + bx + C$,
 where A, B, C are arbitrary constants. (1)

Differentiating (i) successively w. r. t. x we get

$$\frac{dy}{dx} = 2Ax + B$$

$$\frac{d^2y}{dx^2} = 2A$$

$$\frac{d^3y}{dx^3} = 0, \text{ this is the required differential equation of the system of parabolas.}$$

(ii) Since the circles, having their centres on the x-axis, pass through the origin, if $(a, 0)$ be the centre of any member of the family of circles, its radius will be ' a '.

The equation of the circle is

$$(x - a)^2 + (y - 0)^2 = a^2$$

$$\text{i.e., } x^2 + y^2 - 2ax = 0$$

where a is a parameter. (1)

Differentiating w. r. t. x ,

$$2x + 2y \frac{dy}{dx} - 2a = 0$$

$$\text{or, } 2a = 2x + 2y \frac{dy}{dx}$$

(2)

Eliminating 'a' between (1) and (2), we get

$$x^2 + y^2 - \left(2x + 2y \frac{dy}{dx} \right) x = 0$$

$$\text{or,} \quad y^2 - x^2 - 2xy \frac{dy}{dx} = 0$$

is the required differential equation of the circles.

(iii) Let (α, β) be the centre of any circle of the system with constant radius 'a'.

Equation of the circle is

$$(x - \alpha)^2 + (y - \beta)^2 = a^2 \quad \dots (1)$$

where 'a' is constant and α, β are parameters.

Differentiating both sides of (1) w. r. t. x,

$$(x - \alpha) + (y - \beta) \frac{dy}{dx} = 0$$

$$\text{or,} \quad \frac{dy}{dx} = - \frac{x - \alpha}{y - \beta} \quad \dots (2)$$

Differentiating (2) w. r. t. x

$$\begin{aligned} \frac{d^2y}{dx^2} &= - \frac{1 \cdot (y - \beta) - (x - \alpha) \frac{dy}{dx}}{(y - \beta)^2} \\ &= - \frac{(y - \beta)^2 - (x - \alpha)^2}{(y - \beta)^3} \quad [\text{using (2)}] \end{aligned}$$

$$= - \frac{a^2}{(y - \beta)^3} \quad [\text{from (1)}]$$

$$\therefore (y - \beta)^3 = - \frac{a^2}{y_2} \quad \dots (3)$$

$$\text{where } y_2 = \frac{d^2y}{dx^2}.$$

$$\text{Again, } 1 + y_1^2 = 1 + \left(\frac{dy}{dx} \right)^2 = \frac{a^2}{(y - \beta)^2} \quad [\text{from (1)}]$$

$$\text{or, } (y - \beta)^2 = \frac{a^2}{1 + y_1^2} \quad \dots \quad (4)$$

From (3) and (4)

$$\frac{a^4}{y_2^2} = \frac{a^6}{(1 + y_1^2)^3}$$

or, $\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^3 = a^2 \left(\frac{d^2y}{dx^2}\right)^2$ is the required differential equation.

(iv) Let, 'a' be the constant radius of all members of the system of circles and $(\alpha, 0)$ be the centre of a particular member of the system : The equation of the circle is

$$(x - \alpha)^2 + y^2 = a^2 \quad \dots \quad (1)$$

where 'a' is constant and α is a parameter.

Differentiating both sides of (1) w. r. t. x,

$$2(x - \alpha) + 2y \frac{dy}{dx} = 0$$

$$\text{or, } y \frac{dy}{dx} = -(x - \alpha) \quad \dots \quad (2)$$

Eliminating α between (1) and (2) we get differential equation of the system of circles as $\left\{1 + \left(\frac{dy}{dx}\right)^2\right\} y^2 = a^2$.

EXAMPLES XV

1. Show that the differential equation of a system of concentric circles having the centres at the origin is $x dx + y dy = 0$. Interpret the result geometrically.
2. Prove that the differential equation of all circles touching the x-axis at the origin is $(x^2 - y^2) dy - 2xy dx = 0$.
3. (i) Show that the differential equation of all parabolas having their axes parallel to the y-axis is $y_3 = 0$;

(b) with foci at the origin and axes along the x -axis is

$$yy_1^2 + 2xy_1 - y = 0;$$

(ii) Show that the differential equation of the family of circles $x^2 + y^2 + 2gx + 2fy + c = 0$ is $(1 + y_1^2) y_3 - 3y_1 y_2^2 = 0$;

(iii) Show that the differential equation of the family of cardioids $r = a(1 + \cos \theta)$ is $(1 + \cos \theta) dr + r \sin \theta d\theta = 0$.

4. Show that the differential equation of the system of rectangular hyperbolas $xy = c^2$ is $x dy + y dx = 0$, and interpret the result geometrically; deduce that the tangent intercepted between the axes is bisected at the point of contact.
5. Verify that $y + x + 1 = 0$ is a solution of the differential equation $(y - x) dy - (y^2 - x^2) dx = 0$.

6. Show that $V = \frac{A}{r} + B$ is a solution of the differential equation

$$\frac{d^2V}{dr^2} + \frac{2}{r} \frac{dV}{dr} = 0.$$

7. Find the differential equation from the relation

(i) $y = A \sin x + B \cos x + x \sin x$;

(ii) $y = Ae^x + Be^{-x}$;

(iii) $y = A \cos x + B \sin x + C \cosh x + D \sinh x$, where A, B, C, D are arbitrary constants.

8. Eliminate a and b from each of the relations

(i) $y = a \log x + b$;

(ii) $xy = ae^x + be^{-x}$;

(iii) $ax^2 + by^2 = 1$; [C. P. 1945]

(iv) $r = a + b \cos \theta$.

9. (i) Show that the differential equation, whose general solution is

$$y = e_1 x + c_2 x^2, \text{ is } y = xy_1 - \frac{1}{2} x^2 y_2.$$

(ii) Show that

$y = \cos x, y = \sin x, y = c_1 \cos x, y = c_2 \sin x$ are all solutions of the differential equation $y^2 + y = 0$.

[In (i) and (ii), c_1, c_2 are arbitrary constants.]

10. Show that the differential equations, whose general solutions are

(i) $y = A \sin x + B \cos x$,

[C. P. '88]

(ii) $y = A \sinh x + B \cosh x$,

where A and B are arbitrary constants, are respectively

$$\frac{d^2 y}{dx^2} + y = 0 \text{ and } \frac{d^2 y}{dx^2} - y = 0.$$

ANSWERS

1. The radius vector and the tangent at any point are mutually perpendicular.

4. The radius vector and the tangent at any point are equally inclined to the x -axis.

7. (i) $y + y_2 = 2 \cos x$. (ii) $y_2 - y = 0$.

(iii) $y_4 - y = 0$.

8. (i) $xy_2 + y_1 = 0$. (ii) $xy_2 + 2y_1 = xy$.

(iii) $x(yy_2 + y_1^2) = yy_1$. (iv) $r_2 = r_1 \cot \theta$.